Trigonometric Identities

From the unit circle to the half-angle formulas

Trigonometry handout · Yitian Zhang Definitions Identities Worked examples Notes and cautions

Very little in this handout has to be memorised on its own. Two things are taken from the unit circle: the definition of $\cos\theta$ and $\sin\theta$, and one formula, for $\cos(A-B)$, which is proved from the fact that a rotation does not change distances. Every other identity follows from these in a line or two. Figure 1 is the plan.

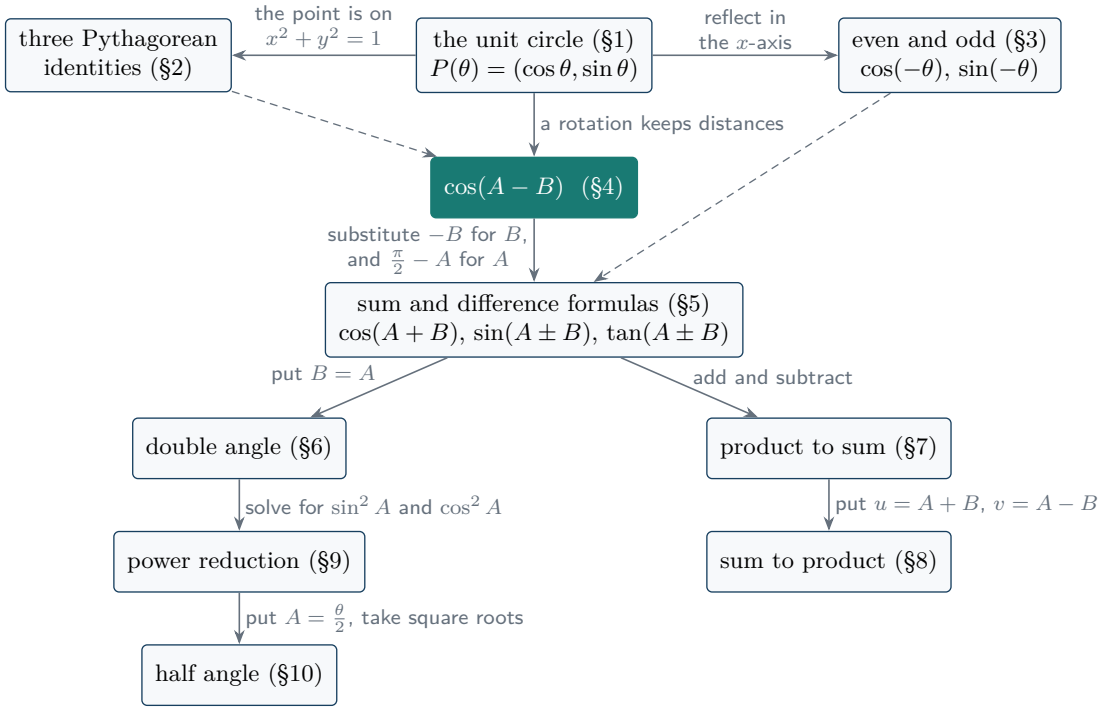


Figure 1. The plan of the handout. Every solid arrow is a derivation of one or two lines. The dashed arrows show where an earlier result is used.

What to memorise

1. The definition: the point at angle θ on the unit circle is $(\cos\theta,\sin\theta)$.
2. $\sin^2\theta+\cos^2\theta=1$.
3. $\cos(A-B)=\cos A\cos B+\sin A\sin B$.

Everything else in this handout can be rebuilt from these three by following the arrows in Figure 1.

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1 The unit circle

Definition (sine and cosine)

The *unit circle* is the circle of radius 1 centred at the origin, $x^2 + y^2 = 1$. Given a real number θ, start at the point $(1, 0)$ and move along the circle through the angle θ, measured in radians: counterclockwise if $\theta > 0$, clockwise if $\theta < 0$. The point where you stop is called $P(\theta)$. Its coordinates are, by definition, the cosine and the sine of θ:

$$P(\theta) = (\cos\theta,\ \sin\theta).$$

For an acute angle this agrees with the right-triangle definition. The triangle in Figure 2 has hypotenuse 1, so its adjacent side is $\cos\theta$ and its opposite side is $\sin\theta$. The unit circle extends the definition to every real number θ.

The other four functions are built from these two, wherever the denominator is not zero:

$$\tan\theta = \frac{\sin\theta}{\cos\theta}, \qquad \cot\theta = \frac{\cos\theta}{\sin\theta},$$
$$\sec\theta = \frac{1}{\cos\theta}, \qquad \csc\theta = \frac{1}{\sin\theta}.$$

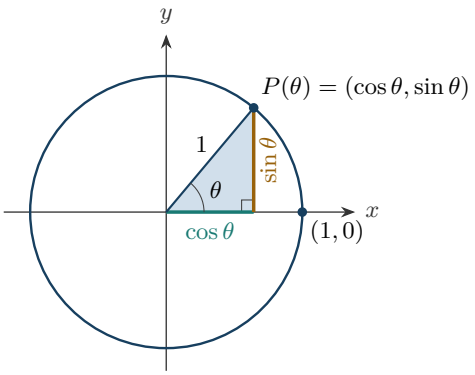


Figure 2. The unit circle.

2 The three Pythagorean identities

Pythagorean identities

$$\sin^2\theta + \cos^2\theta = 1, \qquad \tan^2\theta + 1 = \sec^2\theta, \qquad 1 + \cot^2\theta = \csc^2\theta.$$

Proof

The point $P(\theta) = (\cos\theta, \sin\theta)$ lies on the circle $x^2 + y^2 = 1$, so $\cos^2\theta + \sin^2\theta = 1$.

Divide this identity by $\cos^2\theta$: $\dfrac{\sin^2\theta}{\cos^2\theta} + 1 = \dfrac{1}{\cos^2\theta}$, that is, $\tan^2\theta + 1 = \sec^2\theta$.

Divide the same identity by $\sin^2\theta$: $1 + \dfrac{\cos^2\theta}{\sin^2\theta} = \dfrac{1}{\sin^2\theta}$, that is, $1 + \cot^2\theta = \csc^2\theta$. □

Remember one, derive two

Only the first identity needs to be memorised: the proof shows how to get the other two from it. All three are used most often in their rearranged forms

$$\sin^2\theta = 1 - \cos^2\theta, \qquad \cos^2\theta = 1 - \sin^2\theta, \qquad \sec^2\theta - 1 = \tan^2\theta, \qquad \csc^2\theta - 1 = \cot^2\theta.$$

In integration they are what makes the three trigonometric substitutions work: $x = a\sin\theta$ for $\sqrt{a^2 - x^2}$, $x = a\tan\theta$ for $\sqrt{a^2 + x^2}$, and $x = a\sec\theta$ for $\sqrt{x^2 - a^2}$.

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3 Even and odd

Reflect the plane in the x-axis. A point (x, y) goes to $(x, -y)$, and the point at angle θ goes to the point at angle $-\theta$ (Figure 3). So $P(-\theta) = (\cos\theta, -\sin\theta)$. Reading off the two coordinates, and then dividing them:

Even and odd identities

$$\cos(-\theta) = \cos\theta \qquad \text{(cosine is even)}$$
$$\sin(-\theta) = -\sin\theta \qquad \text{(sine is odd)}$$
$$\tan(-\theta) = -\tan\theta \qquad \text{(tangent is odd)}$$

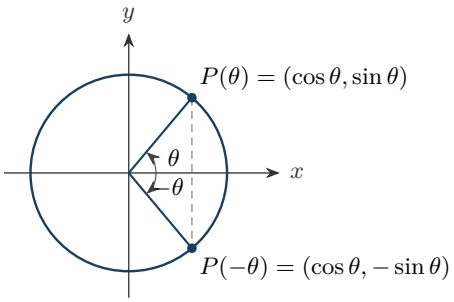


Figure 3. Mirror images in the x-axis.

4 The key formula: $\cos(A - B)$

This is the only formula in the handout that needs a real proof, and the proof needs one geometric fact.

A rotation is an isometry

Rotating the plane about the origin moves the points, but it does not change the distance between any two of them. On the unit circle, the rotation through an angle φ sends the point $P(\theta)$ to the point $P(\theta + \varphi)$.

Theorem (cosine of a difference)

For all real numbers A and B,

$$\cos(A - B) = \cos A \cos B + \sin A \sin B.$$

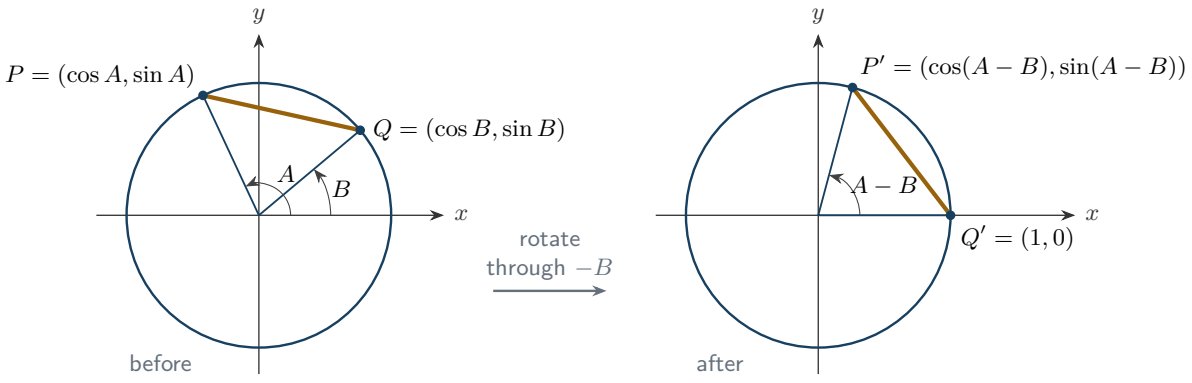


Figure 4. The rotation through $-B$ carries the chord PQ onto the chord $P'Q'$, so the two chords have the same length.

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Proof

Step 1. Mark the points $P = P(A) = (\cos A, \sin A)$ and $Q = P(B) = (\cos B, \sin B)$ on the unit circle. By the distance formula,

$$\begin{aligned} |PQ|^2 &= (\cos A - \cos B)^2 + (\sin A - \sin B)^2 \\ &= \underbrace{\cos^2 A + \sin^2 A}_{1} + \underbrace{\cos^2 B + \sin^2 B}_{1} - 2\,(\cos A \cos B + \sin A \sin B) \\ &= 2 - 2\,(\cos A \cos B + \sin A \sin B). \end{aligned}$$

Step 2. Rotate the plane about the origin through the angle $-B$. Every point of the unit circle stays on the unit circle, and its angle decreases by B. So Q moves to $Q' = P(0) = (1, 0)$, and P moves to $P' = P(A - B) = (\cos(A - B), \sin(A - B))$. By the distance formula again,

$$\begin{aligned} |P'Q'|^2 &= \big(\cos(A - B) - 1\big)^2 + \sin^2(A - B) \\ &= \underbrace{\cos^2(A - B) + \sin^2(A - B)}_{1} + 1 - 2\cos(A - B) = 2 - 2\cos(A - B). \end{aligned}$$

Step 3. A rotation is an isometry, so $|P'Q'| = |PQ|$. Therefore

$$2 - 2\cos(A - B) = 2 - 2\,(\cos A \cos B + \sin A \sin B),$$

and cancelling the 2 and dividing by -2 gives $\cos(A - B) = \cos A \cos B + \sin A \sin B$. □

Why this proof is the one to know

It uses only the distance formula, the identity $\sin^2\theta + \cos^2\theta = 1$, and the fact that a rotation keeps distances. It never draws a triangle with an angle $A - B$ in it, so it never has to ask whether A is larger than B, or which quadrants the angles are in. One argument covers all real A and B.

5 Sum and difference formulas

Everything from here on comes out of the theorem by substitution.

Derivations

Cosine of a sum. Write $A + B = A - (-B)$, apply the theorem, then use the even and odd identities:

$$\cos(A + B) = \cos A \cos(-B) + \sin A \sin(-B) = \cos A \cos B - \sin A \sin B.$$

Cofunction identities. Put $A = \frac{\pi}{2}$ in the theorem. Since $\cos\frac{\pi}{2} = 0$ and $\sin\frac{\pi}{2} = 1$,

$$\cos\Big(\frac{\pi}{2} - B\Big) = 0 \cdot \cos B + 1 \cdot \sin B = \sin B.$$

Replacing B by $\frac{\pi}{2} - B$ in this gives $\cos B = \sin\big(\frac{\pi}{2} - B\big)$. So for every θ,

$$\sin\theta = \cos\Big(\frac{\pi}{2} - \theta\Big), \qquad \cos\theta = \sin\Big(\frac{\pi}{2} - \theta\Big).$$

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Sine of a sum. Turn the sine into a cofunction identity, apply the theorem, and turn back:

$$\begin{aligned}\sin(A+B) &= \cos\Big(\frac{\pi}{2} - (A+B)\Big) = \cos\Big(\Big(\frac{\pi}{2} - A\Big) - B\Big) \\ &= \cos\Big(\frac{\pi}{2} - A\Big)\cos B + \sin\Big(\frac{\pi}{2} - A\Big)\sin B = \sin A\cos B + \cos A\sin B.\end{aligned}$$

Sine of a difference. Replace B by $-B$ in the line above: $\sin(A-B) = \sin A\cos B - \cos A\sin B$.

Tangent. Divide, then divide the top and the bottom by $\cos A\cos B$:

$$\tan(A+B) = \frac{\sin A\cos B + \cos A\sin B}{\cos A\cos B - \sin A\sin B} = \frac{\tan A + \tan B}{1 - \tan A\tan B}.$$

Replacing B by $-B$ gives the formula for $\tan(A-B)$. □

Sum and difference formulas

$$\begin{aligned}\sin(A \pm B) &= \sin A\cos B \pm \cos A\sin B \\ \cos(A \pm B) &= \cos A\cos B \mp \sin A\sin B \\ \tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A\tan B}\end{aligned}$$

Read the upper signs together and the lower signs together.

A way to remember them, and a caution

Sine keeps the sign and mixes the functions: sin cos, cos sin. Cosine changes the sign and keeps the functions apart: cos cos, sin sin.

The sine of a sum is not the sum of the sines: $\sin(A+B) \neq \sin A + \sin B$. For example $\sin(30^\circ + 60^\circ) = 1$, but $\sin 30^\circ + \sin 60^\circ \approx 1.37$.

6 Double-angle formulas

Put $B = A$ in the three sum formulas. For the cosine, the Pythagorean identity gives two more forms: replace $\sin^2 A$ by $1 - \cos^2 A$, or replace $\cos^2 A$ by $1 - \sin^2 A$.

Double-angle formulas

$$\begin{aligned}\sin 2A &= 2\sin A\cos A \\ \cos 2A &= \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A \\ \tan 2A &= \frac{2\tan A}{1 - \tan^2 A}\end{aligned}$$

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7 Product-to-sum formulas

Add and subtract the sum and difference formulas in pairs. The unwanted terms cancel:

$$\cos(A-B)+\cos(A+B)=2\cos A\cos B, \qquad \sin(A+B)+\sin(A-B)=2\sin A\cos B,$$
$$\cos(A-B)-\cos(A+B)=2\sin A\sin B, \qquad \sin(A+B)-\sin(A-B)=2\cos A\sin B.$$

Product-to-sum formulas: the four lines above, divided by 2

$$\cos A\cos B=\tfrac{1}{2}[\cos(A-B)+\cos(A+B)] \qquad \sin A\cos B=\tfrac{1}{2}[\sin(A+B)+\sin(A-B)]$$
$$\sin A\sin B=\tfrac{1}{2}[\cos(A-B)-\cos(A+B)] \qquad \cos A\sin B=\tfrac{1}{2}[\sin(A+B)-\sin(A-B)]$$

They turn a product, which is hard to integrate, into a sum, which is easy (Example 5).

8 Sum-to-product formulas

Read the four equations at the start of the last section from left to right, and rename the angles. Put

$$u=A+B, \quad v=A-B, \qquad \text{so that} \qquad A=\frac{u+v}{2}, \quad B=\frac{u-v}{2}.$$

Sum-to-product formulas

$$\sin u+\sin v=2\sin\frac{u+v}{2}\cos\frac{u-v}{2} \qquad \cos u+\cos v=2\cos\frac{u+v}{2}\cos\frac{u-v}{2}$$
$$\sin u-\sin v=2\cos\frac{u+v}{2}\sin\frac{u-v}{2} \qquad \cos u-\cos v=-2\sin\frac{u+v}{2}\sin\frac{u-v}{2}$$

Note the minus sign in the last one: it comes from $\cos(A+B)-\cos(A-B)=-2\sin A\sin B$. These formulas turn a sum into a product, which is what is needed to factor an expression, to solve an equation, or to simplify a quotient (Examples 6 and 8).

9 Power-reduction formulas

Solve $\cos 2A=1-2\sin^2 A$ for $\sin^2 A$, and solve $\cos 2A=2\cos^2 A-1$ for $\cos^2 A$. Divide the two results to get $\tan^2 A$.

Power-reduction formulas

$$\sin^2 A=\frac{1-\cos 2A}{2}, \qquad \cos^2 A=\frac{1+\cos 2A}{2}, \qquad \tan^2 A=\frac{1-\cos 2A}{1+\cos 2A}.$$

They lower the power from 2 to 1, at the price of doubling the angle. This is how $\sin^2 x$ and $\cos^2 x$ are integrated (Example 4). For a fourth power, use them twice:

$$\cos^4 x=\tfrac{1}{4}(1+\cos 2x)^2=\tfrac{1}{4}(1+2\cos 2x+\cos^2 2x)=\tfrac{1}{8}(3+4\cos 2x+\cos 4x).$$

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10 Half-angle formulas

Put $A = \frac{\theta}{2}$ in the power-reduction formulas, so that $2A = \theta$, and take square roots.

Half-angle formulas

$$\sin\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{2}}, \qquad \cos\frac{\theta}{2} = \pm\sqrt{\frac{1+\cos\theta}{2}}, \qquad \tan\frac{\theta}{2} = \pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}.$$

$$\tan\frac{\theta}{2} = \frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta} \qquad \text{(no sign to choose)}.$$

The sign is not a free choice

The sign $\pm$ is decided by the quadrant in which $\frac{\theta}{2}$ lies, not the quadrant of θ. For example, if $\pi < \theta < 2\pi$ then $\frac{\pi}{2} < \frac{\theta}{2} < \pi$, which is the second quadrant, so $\sin\frac{\theta}{2}$ is positive and $\cos\frac{\theta}{2}$ is negative.

The two forms of the tangent without a sign

Multiply the top and the bottom of $\tan\frac{\theta}{2} = \frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}$ by $2\cos\frac{\theta}{2}$ for the first form, and by $2\sin\frac{\theta}{2}$ for the second. Then use $\sin 2A = 2\sin A\cos A$, $2\cos^2 A = 1+\cos 2A$ and $2\sin^2 A = 1-\cos 2A$, with $A = \frac{\theta}{2}$:

$$\tan\frac{\theta}{2} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \frac{\sin\theta}{1+\cos\theta}, \qquad \tan\frac{\theta}{2} = \frac{2\sin^2\frac{\theta}{2}}{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \frac{1-\cos\theta}{\sin\theta}. \qquad \square$$

11 Worked examples

Example 1 ▪ an exact value — difference formula

Find the exact value of $\cos 15^\circ$.

Solution. Write $15^\circ = 45^\circ - 30^\circ$.

$$\cos 15^\circ = \cos 45^\circ\cos 30^\circ + \sin 45^\circ\sin 30^\circ = \frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\cdot\frac{1}{2} = \boxed{\frac{\sqrt{6}+\sqrt{2}}{4}}.$$

Example 2 ▪ another exact value — half angle

Find the exact value of $\sin\frac{\pi}{8}$.

Solution. $\frac{\pi}{8}$ is half of $\frac{\pi}{4}$ and lies in the first quadrant, so the sign is $+$.

$$\sin\frac{\pi}{8} = \sqrt{\frac{1-\cos\frac{\pi}{4}}{2}} = \sqrt{\frac{1-\frac{\sqrt{2}}{2}}{2}} = \sqrt{\frac{2-\sqrt{2}}{4}} = \boxed{\frac{\sqrt{2-\sqrt{2}}}{2}}.$$

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Example 3 ▪ from one value to many Pythagorean, double angle, half angle

Given $\sin\theta = \frac{3}{5}$ with $\frac{\pi}{2} < \theta < \pi$ (the second quadrant), find $\cos\theta$, $\sin 2\theta$, $\cos 2\theta$ and $\cos\frac{\theta}{2}$.

Solution. **Pythagorean identity.** $\cos^2\theta = 1 - \sin^2\theta = 1 - \frac{9}{25} = \frac{16}{25}$. Cosine is negative in the second quadrant, so $\cos\theta = \boxed{-\frac{4}{5}}$.

Double angle.

$$\sin 2\theta = 2\sin\theta\cos\theta = 2\cdot\frac{3}{5}\cdot\left(-\frac{4}{5}\right) = \boxed{-\frac{24}{25}}, \qquad \cos 2\theta = 1 - 2\sin^2\theta = 1 - \frac{18}{25} = \boxed{\frac{7}{25}}.$$

Half angle. Halving $\frac{\pi}{2} < \theta < \pi$ gives $\frac{\pi}{4} < \frac{\theta}{2} < \frac{\pi}{2}$: the half angle is in the first quadrant, so its cosine is positive.

$$\cos\frac{\theta}{2} = +\sqrt{\frac{1+\cos\theta}{2}} = \sqrt{\frac{1-\frac{4}{5}}{2}} = \sqrt{\frac{1}{10}} = \boxed{\frac{\sqrt{10}}{10}}.$$

Example 4 ▪ integrating even powers power reduction

Find (a) $\int \sin^2 x\,\mathrm{d}x$ and (b) $\int \sin^2 x\cos^2 x\,\mathrm{d}x$.

Solution. (a) Lower the power first:

$$\int \sin^2 x\,\mathrm{d}x = \int \frac{1-\cos 2x}{2}\,\mathrm{d}x = \boxed{\frac{x}{2} - \frac{\sin 2x}{4} + C}.$$

(b) By the double-angle formula, $\sin x\cos x = \frac{1}{2}\sin 2x$, so $\sin^2 x\cos^2 x = \frac{1}{4}\sin^2 2x$. Now lower the power, with $A = 2x$:

$$\int \sin^2 x\cos^2 x\,\mathrm{d}x = \frac{1}{4}\int \sin^2 2x\,\mathrm{d}x = \frac{1}{4}\int \frac{1-\cos 4x}{2}\,\mathrm{d}x = \boxed{\frac{x}{8} - \frac{\sin 4x}{32} + C}.$$

Example 5 ▪ integrating a product product to sum

Find $\int \sin 5x\cos 3x\,\mathrm{d}x$.

Solution. Use $\sin A\cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$ with $A = 5x$ and $B = 3x$:

$$\int \sin 5x\cos 3x\,\mathrm{d}x = \frac{1}{2}\int (\sin 8x + \sin 2x)\,\mathrm{d}x = \boxed{-\frac{\cos 8x}{16} - \frac{\cos 2x}{4} + C}.$$

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Example 6 ▪ simplifying a quotient sum to product

Show that $\dfrac{\sin 3x + \sin x}{\cos 3x + \cos x} = \tan 2x$.

Solution. Use the sum-to-product formulas with $u = 3x$ and $v = x$, so that $\frac{u+v}{2} = 2x$ and $\frac{u-v}{2} = x$:

$$\frac{\sin 3x + \sin x}{\cos 3x + \cos x} = \frac{2\sin 2x \cos x}{2\cos 2x \cos x} = \frac{\sin 2x}{\cos 2x} = \tan 2x,$$

wherever the left side is defined.

Example 7 ▪ a triple-angle formula sum formula, double angle

Show that $\cos 3A = 4\cos^3 A - 3\cos A$.

Solution. Write $3A = 2A + A$ and use the sum formula, then the double-angle formulas.

$$\begin{aligned}\cos 3A &= \cos 2A \cos A - \sin 2A \sin A \\ &= (2\cos^2 A - 1)\cos A - (2\sin A \cos A)\sin A \\ &= 2\cos^3 A - \cos A - 2(1 - \cos^2 A)\cos A \\ &= 4\cos^3 A - 3\cos A.\end{aligned}$$

The form $\cos 2A = 2\cos^2 A - 1$ was chosen because the answer is wanted in terms of $\cos A$ only, and $\sin^2 A$ was replaced by $1 - \cos^2 A$ for the same reason.

Example 8 ▪ solving an equation sum to product

Solve $\sin 3x + \sin x = 0$ for $0 \le x < 2\pi$.

Solution. A sum equal to zero is hard to solve, but a product equal to zero is easy. By the sum-to-product formula with $u = 3x$ and $v = x$, the equation becomes

$$2\sin 2x \cos x = 0, \qquad \text{so} \qquad \sin 2x = 0 \quad \text{or} \quad \cos x = 0.$$

- $\sin 2x = 0$ with $0 \le 2x < 4\pi$ gives $2x = 0,\ \pi,\ 2\pi,\ 3\pi$, that is, $x = 0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}$.
- $\cos x = 0$ gives $x = \frac{\pi}{2},\ \frac{3\pi}{2}$, which are already in the list.

The solutions are $\boxed{x = 0,\ \dfrac{\pi}{2},\ \pi,\ \dfrac{3\pi}{2}}$.

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12 Summary sheet

Pythagorean

$$\sin^2\theta+\cos^2\theta=1 \qquad \tan^2\theta+1=\sec^2\theta \qquad 1+\cot^2\theta=\csc^2\theta$$

Even and odd

$$\cos(-\theta)=\cos\theta \qquad \sin(-\theta)=-\sin\theta \qquad \tan(-\theta)=-\tan\theta$$

Cofunction

$$\sin\left(\frac{\pi}{2}-\theta\right)=\cos\theta \qquad \cos\left(\frac{\pi}{2}-\theta\right)=\sin\theta$$

Sum and difference

$$\sin(A\pm B)=\sin A\cos B\pm\cos A\sin B \qquad \cos(A\pm B)=\cos A\cos B\mp\sin A\sin B$$

$$\tan(A\pm B)=\frac{\tan A\pm\tan B}{1\mp\tan A\tan B}$$

Double angle

$$\sin 2A=2\sin A\cos A \qquad \tan 2A=\frac{2\tan A}{1-\tan^2 A}$$

$$\cos 2A=\cos^2 A-\sin^2 A=2\cos^2 A-1=1-2\sin^2 A$$

Product to sum

$$\cos A\cos B=\tfrac{1}{2}\left[\cos(A-B)+\cos(A+B)\right] \qquad \sin A\cos B=\tfrac{1}{2}\left[\sin(A+B)+\sin(A-B)\right]$$

$$\sin A\sin B=\tfrac{1}{2}\left[\cos(A-B)-\cos(A+B)\right] \qquad \cos A\sin B=\tfrac{1}{2}\left[\sin(A+B)-\sin(A-B)\right]$$

Sum to product

$$\sin u+\sin v=2\sin\frac{u+v}{2}\cos\frac{u-v}{2} \qquad \cos u+\cos v=2\cos\frac{u+v}{2}\cos\frac{u-v}{2}$$

$$\sin u-\sin v=2\cos\frac{u+v}{2}\sin\frac{u-v}{2} \qquad \cos u-\cos v=-2\sin\frac{u+v}{2}\sin\frac{u-v}{2}$$

Power reduction

$$\sin^2 A=\frac{1-\cos 2A}{2} \qquad \cos^2 A=\frac{1+\cos 2A}{2} \qquad \tan^2 A=\frac{1-\cos 2A}{1+\cos 2A}$$

Half angle

$$\sin\frac{\theta}{2}=\pm\sqrt{\frac{1-\cos\theta}{2}} \qquad \cos\frac{\theta}{2}=\pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\frac{\theta}{2}=\pm\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}=\frac{\sin\theta}{1+\cos\theta}=\frac{1-\cos\theta}{\sin\theta}$$

Exact values

θ	0	$\frac{\pi}{6}=30^\circ$	$\frac{\pi}{4}=45^\circ$	$\frac{\pi}{3}=60^\circ$	$\frac{\pi}{2}=90^\circ$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	not defined

In the half-angle formulas the sign is decided by the quadrant of $\frac{\theta}{2}$.

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13 Practice

1. Find the exact value of $\sin 75°$.
2. Find the exact value of $\tan 15°$.
3. Find the exact value of $\cos 105°$.
4. Given $\cos\theta = -\frac{5}{13}$ with $\pi < \theta < \frac{3\pi}{2}$ (the third quadrant), find $\sin 2\theta$, $\cos 2\theta$ and $\cos\frac{\theta}{2}$.
5. Show that $\sin 3A = 3\sin A - 4\sin^3 A$.
6. Find $\displaystyle\int \cos^2 3x \,\mathrm{d}x$.
7. Find $\displaystyle\int \sin 2x \sin 4x \,\mathrm{d}x$.
8. Write $\sin 5x - \sin x$ as a product.
9. Show that $\dfrac{1-\cos 2x}{\sin 2x} = \tan x$.
10. Find $\displaystyle\int \cos^4 x \,\mathrm{d}x$.
11. Solve $\cos 3x + \cos x = 0$ for $0 \le x < 2\pi$.

Answers

1. $\sin 75° = \sin(45° + 30°) = \dfrac{\sqrt{6}+\sqrt{2}}{4}$.
2. $\tan 15° = \tan(45° - 30°) = \dfrac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \dfrac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3}$.
3. $\cos 105° = \cos(60° + 45°) = \dfrac{\sqrt{2}-\sqrt{6}}{4}$.
4. $\sin\theta = -\frac{12}{13}$, so $\sin 2\theta = \frac{120}{169}$ and $\cos 2\theta = -\frac{119}{169}$. Halving $\pi < \theta < \frac{3\pi}{2}$ puts the half angle in the second quadrant, so $\cos\frac{\theta}{2} = -\sqrt{\frac{1-\frac{5}{13}}{2}} = -\dfrac{2\sqrt{13}}{13}$.
5. $\sin 3A = \sin(2A + A) = \sin 2A\cos A + \cos 2A \sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A$. Replace $\cos^2 A$ by $1 - \sin^2 A$ and collect terms.
6. $\dfrac{x}{2} + \dfrac{\sin 6x}{12} + C$.
7. $\dfrac{1}{2}\displaystyle\int (\cos 2x - \cos 6x)\,\mathrm{d}x = \dfrac{\sin 2x}{4} - \dfrac{\sin 6x}{12} + C$.
8. $2\cos 3x \sin 2x$.
9. $\dfrac{1-\cos 2x}{\sin 2x} = \dfrac{2\sin^2 x}{2\sin x\cos x} = \tan x$.
10. $\dfrac{3x}{8} + \dfrac{\sin 2x}{4} + \dfrac{\sin 4x}{32} + C$, from $\cos^4 x = \frac{1}{8}(3 + 4\cos 2x + \cos 4x)$.
11. $2\cos 2x\cos x = 0$, so $\cos 2x = 0$ or $\cos x = 0$: $\quad x = \dfrac{\pi}{4}, \dfrac{\pi}{2}, \dfrac{3\pi}{4}, \dfrac{5\pi}{4}, \dfrac{3\pi}{2}, \dfrac{7\pi}{4}$.

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